25901639

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Second Semester BCA (Honours)

Web Technologies (MG2SECBCA100) – April 2026

Answer Scheme

Duration: 1 Hour | Maximum Marks: 35

PART A – Short Answer Questions

Answer any 4 questions. Each question carries 5 marks. (4 × 5 = 20 Marks)

1. Implement a navbar with an example

Definition:

A navigation bar (navbar) is a section of a webpage that contains links to other pages or sections of the site. It is usually placed at the top of the page.

Example Code:

```
<!DOCTYPE html>
<html>
<head>
  <style>
    nav { background-color: #333; }
    nav ul { list-style: none; margin: 0; padding: 0; display: flex; }
    nav ul li a {
      display: block; color: white;
      padding: 14px 20px; text-decoration: none;
    }
    nav ul li a:hover { background-color: #555; }
  </style>
</head>
<body>
  <nav>
    <ul>
      <li><a href='#home'>Home</a></li>
      <li><a href='#about'>About</a></li>
      <li><a href='#services'>Services</a></li>
      <li><a href='#contact'>Contact</a></li>
    </ul>
  </nav>
</body>
</html>
```

Mark Allocation:

Definition of navbar	2 Marks
Correct use of <nav>, , , <a> tags	3 Marks

2. Define web hosting, and why is it necessary for websites?

Definition:
Web hosting is a service that allows individuals and organizations to store website files on a server connected to the internet, making the website accessible to users worldwide.

Why Web Hosting is Necessary:

- Online Storage: Website files (HTML, CSS, images, databases) need to be stored on a server that is always online.
- 24/7 Accessibility: Hosting ensures users can visit the website at any time from anywhere in the world.
- Domain Linking: Web hosting allows a domain name (e.g., www.example.com) to be linked to the website through DNS.
- Security: Hosting providers offer SSL certificates, firewalls, and malware protection.
- Email Hosting: Business email accounts linked to the domain are also provided.

Types of Web Hosting:

- Shared Hosting – Multiple websites on one server
- VPS Hosting – Dedicated virtual resources
- Dedicated Hosting – Entire server for one website
- Cloud Hosting – Resources spread across multiple servers

Mark Allocation:

Clear definition of web hosting	2 Marks
Any 2 reasons explained clearly	3 Marks

3. Create a definition list of any five input devices.

Definition:
A definition list in HTML is used to display terms and their descriptions. It uses the <dl> tag for the list, <dt> for the term, and <dd> for the description.

Syntax:

```
<dl>
  <dt>Term</dt>
  <dd>Description</dd>
</dl>
```

Example:

```
<!DOCTYPE html>
<html>
<body>
  <h2>Input Devices</h2>
  <dl>
    <dt>Keyboard</dt>
    <dd>Used to type text and commands into the computer.</dd>
    <dt>Mouse</dt>
    <dd>A pointing device used to control the cursor on screen.</dd>
    <dt>Scanner</dt>
    <dd>Converts physical documents into digital format.</dd>
    <dt>Microphone</dt>
```

```
    <dd>Captures audio and converts it to digital signals.</dd>
    <dt>Webcam</dt>
    <dd>Captures video for video calls and recording.</dd>
  </dl>
</body>
</html>
```

Mark Allocation:

Correct explanation of <dl>, <dt>, <dd>	2 Marks
Five input devices listed	3 Marks

4. List and briefly explain the advantages of using AJAX in web applications.

What is AJAX?

AJAX (Asynchronous JavaScript and XML) is a technique that allows web pages to communicate with a server and update content without reloading the entire page.

Advantages:

- Asynchronous Communication: Data is sent and received in the background without interrupting the user's interaction with the page.
- Reduced Server Load: Only the required data is fetched instead of reloading the entire page, reducing bandwidth and server usage.
- Faster User Experience: The page feels faster because only parts of it are updated, not the whole page.
- Improved Interactivity: Features like live search suggestions, auto-complete, and real-time updates are possible.
- Platform Independence: Works with any server-side language (PHP, Python, Java) and any modern browser.
- Partial Page Update: Only specific sections are updated, preserving the rest of the page layout.

Mark Allocation:

Definition of AJAX	1 Mark
Any 4 advantages clearly explained	4 Marks

5. Describe the key features of AJAX-based web applications with examples.

Introduction:

AJAX-based web applications are dynamic applications that update content by communicating with a server asynchronously using the XMLHttpRequest object or the Fetch API, without a full page reload.

Key Features:

- Asynchronous Data Loading: Data loads in the background while the user continues to interact. Example: Gmail loads new emails without refreshing the page.
- Partial Page Updates: Only a specific part of the page is updated. Example: Facebook's newsfeed adds new posts without a full reload.
- Real-Time Data: Applications can fetch live data. Example: Stock price dashboards, live sports scores.
- Reduced Bandwidth: Only JSON or XML data is transferred, not full HTML pages. Example: Google Maps loads map tiles incrementally.
- Better User Experience: Smooth transitions, live form validations, and dynamic content. Example: Checking username availability while signing up.
- Cross-Platform Support: Works across browsers and different server-side technologies.

Basic AJAX Example:

```
const xhr = new XMLHttpRequest();
xhr.open('GET', 'data.php', true);
xhr.onload = function() {
  if (xhr.status === 200) {
    document.getElementById('result').innerHTML = xhr.responseText;
  }
};
xhr.send();
```

Mark Allocation:

Any 3 features with examples	3 Marks
Code example showing AJAX call	2 Marks

6. Implement onchange event in JavaScript

Definition:

The onchange event fires when the value of an HTML element (such as <input>, <select>, or <textarea>) changes and the element loses focus. It is used for dynamic updates and form validations.

Syntax:

```
<input onchange='functionName()'>
```

Example 1 – Text Input:

```
<input type='text' id='name' onchange='showMsg()' placeholder='Enter
name'>
<p id='output'></p>
<script>
  function showMsg() {
    let val = document.getElementById('name').value;
    document.getElementById('output').innerHTML = 'You entered: ' + val;
  }
</script>
```

Example 2 – Dropdown Select:

```
<select id='color' onchange='changeColor()'>
  <option value='red'>Red</option>
  <option value='blue'>Blue</option>
  <option value='green'>Green</option>
</select>
<script>
  function changeColor() {
    let c = document.getElementById('color').value;
    document.body.style.backgroundColor = c;
  }
</script>
```

Mark Allocation:

Syntax shown correctly	2.5 Marks
Working code example	2.5 Marks

PART B – Essay Questions

Answer any 1 question. Each question carries 15 marks. (1 × 15 = 15 Marks)

7. Describe the purpose and working of different web protocols.

Introduction:

A web protocol is a set of rules that defines how data is transmitted and received over the internet. Protocols ensure reliable communication between different devices and systems.

1. HTTP – HyperText Transfer Protocol

- Purpose: The foundation of data communication on the World Wide Web.
- Port: 80
- Working: The browser (client) sends an HTTP request to the web server. The server processes it and responds with the requested resource (HTML page, image, etc.).
- Methods: GET (retrieve), POST (submit), PUT (update), DELETE (remove).
- It is a stateless protocol – each request is independent.

```
Request:  GET /index.html HTTP/1.1  Host: www.example.com
Response: HTTP/1.1 200 OK           Content-Type: text/html
```

2. HTTPS – HTTP Secure

- Purpose: Encrypted version of HTTP for secure data transfer.
- Port: 443
- Working: Uses SSL/TLS certificates to encrypt communication between client and server.
- Used for online banking, login forms, and e-commerce to protect sensitive data.

3. FTP – File Transfer Protocol

- Purpose: Used to transfer files between a client and a server.
- Ports: 21 (command), 20 (data)
- Working: Client connects using credentials and uploads or downloads files.
- Used by web developers to upload website files to a hosting server.
- Variants: FTPS (FTP Secure), SFTP (SSH File Transfer Protocol).

4. SMTP – Simple Mail Transfer Protocol

- Purpose: Used for sending email messages.
- Port: 25 (or 465/587 for secure connections)
- Working: The mail client sends the email to the SMTP server, which routes it to the recipient's mail server.

5. POP3 – Post Office Protocol version 3

- Purpose: Used to retrieve emails from the server to a local client.
- Port: 110
- Working: Downloads emails to the local device and (by default) deletes them from the server.
- Limitation: Emails are stored locally and are not accessible from multiple devices.

6. IMAP – Internet Message Access Protocol

- Purpose: Allows email to be accessed and managed directly on the mail server.
- Port: 143
- Working: Emails remain on the server and are synced across multiple devices.
- Advantage over POP3: Accessible from multiple devices simultaneously.

7. DNS – Domain Name System

- Purpose: Translates human-readable domain names into IP addresses.
- Port: 53
- Working: When you type www.google.com, a DNS query is sent to a DNS resolver, which finds the corresponding IP address (e.g., 142.250.195.46) and returns it to the browser.
- Common DNS Records: A (IPv4 address), AAAA (IPv6), MX (mail server), CNAME (alias).

8. TCP/IP – Transmission Control Protocol / Internet Protocol

- Purpose: The core communication protocol suite of the internet.
- TCP: Ensures reliable, ordered delivery of data packets.
- IP: Handles addressing and routing of packets across networks.
- Working: Data is split into packets, each labelled with source and destination IP. TCP ensures all packets arrive and are reassembled in order.

Mark Allocation:

HTTP – purpose, working, methods	2 Marks
HTTPS – difference from HTTP, SSL/TLS	2 Marks
FTP – purpose, ports, use case	2 Marks
SMTP / POP3 / IMAP – email protocols	3 Marks
DNS – domain resolution process	3 Marks
TCP/IP – role in internet communication	3 Marks

8. Discuss the different string manipulation methods in JavaScript. Explain with examples, definition and syntax.

Introduction:

In JavaScript, a string is a sequence of characters. The String object provides built-in methods to search, manipulate, and transform strings. Strings are immutable – methods return a new string without changing the original.

1. length

Returns the number of characters in a string.

```
let str = 'Hello';
console.log(str.length);  // 5
```

2. toUpperCase() and toLowerCase()

Converts all characters to uppercase or lowercase.

```
let s = 'JavaScript';
console.log(s.toUpperCase());  // JAVASCRIPT
console.log(s.toLowerCase());  // javascript
```

3. charAt(index)

Returns the character at the specified index (starts at 0).

```
let s = 'MGU';
console.log(s.charAt(0));  // M
console.log(s.charAt(2));  // U
```

4. indexOf() and lastIndexOf()

indexOf() returns the position of the first occurrence of a substring. lastIndexOf() returns the last. Returns -1 if not found.

```
let str = 'banana';
```

```
console.log(str.indexOf('a'));      // 1
console.log(str.lastIndexOf('a'));  // 5
```

5. slice(start, end)

Extracts part of a string from start index to end index (end not included). Supports negative indices.

```
let str = 'Web Technologies';
console.log(str.slice(0, 3));  // Web
console.log(str.slice(-12));   // Technologies
```

6. substring(start, end)

Similar to slice but does not support negative indices.

```
let str = 'JavaScript';
console.log(str.substring(0, 4));  // Java
console.log(str.substring(4));     // Script
```

7. replace() and replaceAll()

replace() replaces the first occurrence of a substring. replaceAll() replaces all occurrences.

```
let str = 'Hello World';
console.log(str.replace('World', 'MGU'));  // Hello MGU
let s = 'aabbaa';
console.log(s.replaceAll('a', 'x'));       // xxbbxx
```

8. split(separator)

Splits a string into an array of substrings based on a separator.

```
let csv = 'red,green,blue';
let arr = csv.split(',');
console.log(arr);  // ['red', 'green', 'blue']
```

9. trim(), trimStart(), trimEnd()

Removes whitespace from both ends, the start only, or the end only.

```
let s = '   Hello   ';
console.log(s.trim());       // 'Hello'
console.log(s.trimStart()); // 'Hello   '
```

10. includes(), startsWith(), endsWith()

Check whether a string contains, starts with, or ends with a given substring. Returns true or false.

```
let str = 'JavaScript is fun';
console.log(str.includes('fun'));     // true
console.log(str.startsWith('Java')); // true
console.log(str.endsWith('fun'));    // true
```

11. concat()

Joins two or more strings together.

```
let a = 'Web', b = ' Technologies';
console.log(a.concat(b));  // Web Technologies
```

12. repeat(n)

Returns a new string repeated n times.

```
let s = 'Hi! ';
console.log(s.repeat(3));  // Hi! Hi! Hi!
```

13. padStart() and padEnd()

Pads the beginning or end of a string with a given character until it reaches the specified length.

```
let num = '7';
console.log(num.padStart(4, '0'));  // 0007
console.log(num.padEnd(4, '*'));    // 7***
```

Mark Allocation:

length, toUpperCase(), toLowerCase() with examples	3 Marks
charAt(), indexOf(), lastIndexOf()	3 Marks
slice(), substring()	2 Marks
replace(), replaceAll()	2 Marks
split(), trim()	2 Marks
includes(), startsWith(), endsWith()	3 Marks

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